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PLC Programming Instructions

Programmable Logic Controllers

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Agenda

Data Types

Basic Programming Instructions

Timers and Counters

Advanced Programming Instructions

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Data Types

BIT

- 1 bit
- The smallest unit of memory
- Represents discrete binary states
  - 0 (OFF/FALSE)
  - 1 (ON/TRUE)

NIBBLE

- 4 bits
- 1 binary coded decimal (BCD) digit

BYTE

- 8 Bits
- 256 unique combinations
- Typically used for ASCII characters–1 BYTE = 1 character

WORD

- 16 Bits
- Historical standard register size in PLCs (AB SLC 500/Siemens S7-300)

DWORD/LWORD

- DWORD–32 bits
- LWORD–64 bits
- Modern architecture
- Used for high precision motion/math in older systems

Sub-Bit Access

- These data types are most often used as a number
- But they are *also* just a collection of bits
- We can access the individual bit inside a word using *dot notation*
  - `Word_Var.0`, `Word_Var.7`, etc.

IEC 61131-3 Elementary Types

Type	Keyword	Bits	Signed Range	Common Industrial Usage
Boolean	`BOOL`	1	0 or 1 (FALSE / TRUE)	Proximity sensors, pushbuttons, coils
Short Integer	`SINT`	8	-128 to +127	Small part counts, status flags
Integer	`INT`	16	-32,768 to +32,767	Analog inputs (0-10V, 4-20mA raw counts)
Double Integer	`DINT`	32	-2,147,483,648 to +2,147,483,647	High-speed encoders, totalizer counts
Real (Float)	`REAL`	32	IEEE 754 Floating Point	Scaled engineering units (PSI, GPM, °C)

Signed Integers

- Integers can be signed (+/-) or unsigned
- The defautl in most PLCs is signed–unsigned is optional
- In a signed integer the first bit is the sign: 0 = +, 1 = -
- Because we lose a bit, the capacity of the number is effectively cut in half
- `INT`: -32,768 to 32,767
- `UINT`: 0 to 65,535

Floating Point Numbers

- IEEE 754 Format
- 32-bits: 1 sign, 8 exponent bits, 23 mantissa bits
- An integer is *only* whole numbers, `REAL` must be used if a decimal point is necessary
- *NEVER* use an equality check (`REAL_1 = 10.0`) on a `REAL`–floating point numbers have in-built rounding errors

Addressing Memory in PLCs

- **Direct Memory Addressing:** Logic references explicit *physical* register locations in the CPU
- **Tag-Based Addressing:** Programmer creates descriptive, symbolic names stored in a database
- IEC 61131-3 standardizes both approaches
- Older PLCs pretty exclusively direct addressed, modern PLCs mostly use tag-based databases

Direct vs. Tag-Based Syntax

AB Direct (SLC/Micrologix)
Uses fixed file letters & numbers
`I:1/0`–Input slot 1, bit 0
`B3:0/0`–Binary bit word 0, bit 3
`N7:0`–Integer file 7, register 0

IEC 61131-3 Direct Syntax
Uses standard prefix notation
`%IX0.0`–Input byte 0, bit 0
`%QX0.0`–Output byte 0, bit 0
`%MW100`–Memory word 100

Tag-Based (Logix/TIA)
Uses symbolic variable names
`Conveyor_Start_PB`
`Tank_Level_Scaled`
Hardware mapped vis I/O aliasing

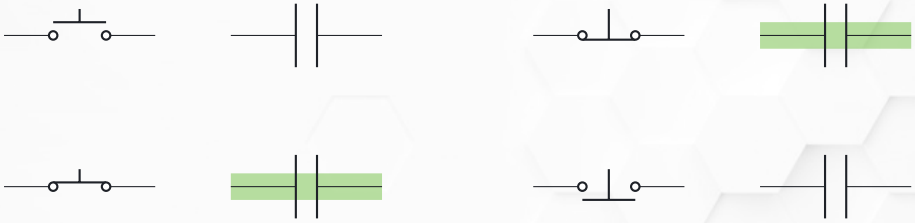
Basic Programming Instructions

The 3 Basic Instructions

- Ladder logic doesn't care what the input *is*
- All that matters is on/off
- Remember, these were intentionally designed to look like relay contacts and coils

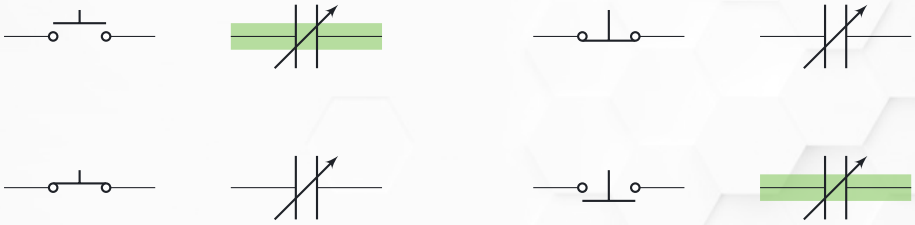
Examine if Closed (XIC)

True when the input is true (on)



Examine if Open (XIO)

True when the input is false (off)

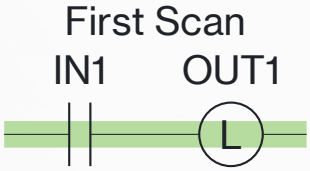


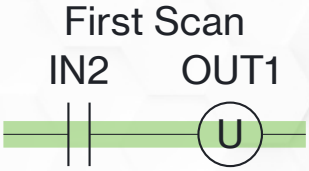
Output Energize (OTE)

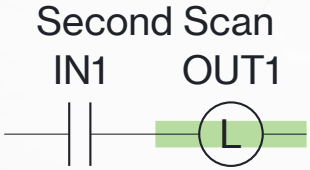
True when the logic before it is true

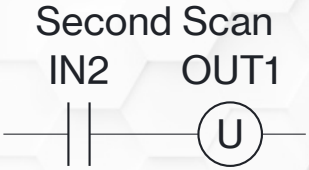
Output Latch (OTL) and Unlatch (OTU)

Once turned on the outputs *stays on* until explcitly unlatched









Take Caution with Latches

- Be *very* careful with latches–my recommendation is to not use them as much as possible
- OTE is *volatile*–when power cycles the output is set to 0 (off)
- OTL is *NOT* volatile–if it's on when power goes out it will be on when power is restored
- This can and has caused unwanted/unsafe machine motion

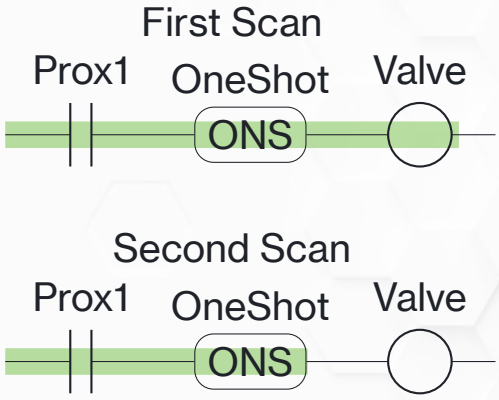
Latch/Unlatch on Key Platforms

How Key Platforms Handle Latch/Unlatch on Power Cycle

Manufacturer	Equivalent Instructions	Power Cycle Behavior	Memory Control Mehcanism
Allen-Bradley (Logix5000)	OTL/OTU	Retains state across power cycles (non-volatile)	Tied directly to the instruction choice (OTL/OTU memory retention vs. OTE non-retention)
Siemens (TIA Portal)	S (Set)/ R (Reset)	Resets to `0` unless declared as Retentive	Bit/DB retain settings in tag table
Beckhoff (TwinCAT)/ CODESYS	S (Set)/ R (Reset)	Resets to `0` unless located in Retain memory	`RETAIN` or `PERSISTENT` keywords
Mitsubishi (GX Works)	SET/RST	Retains state if mapped to Latch Range	Latch (L) device memory range
Omron (Sysmac Studio)	Set/Rst	Resets to `0` unless declared as Retentive	Retain attribute checkbox on VAR

Oneshot

A oneshot is true for *one scan* while the logic before it is true–the logic has to turn false to reset it



Oneshot Variable

- A oneshot variable can only be used *once*–each oneshot instruction needs a unique variable
- This is a good case for using sub-bits
  - Variable: OneShot, Data Type: `DINT`
  - This provides 32 variables in one–`OneShot.0`, `OneShot.1`, etc.

Timers and Counters

Timers and Counters

- Timers time, counters count–but their under-the-hood structure is *very* similar
- “Timer” and “counter” refer to the *variable type*
- The timer or counter variable is then used in conjunction with a timer or counter instruction

Timer and Counter Variable Types

- Timers and counters are made up of 3 words:
  1. Word 0: Control word–used as bits, what the bits *do* vary between timers and counters
  2. Word 1: Preset (PRE)–the time/count target
  3. Word 2: Accumulated (ACC)–the current time/count

Timer Variables

EN Rung logic before the instruction is true
TT The timer is actively timing
DN The preset value has been reached
PRE The target time, in *milliseconds*
ACC The current timer value, in *milliseconds*

Time Base

- Some platforms use a timer time base
- Time base × preset = total time
- AB RSlogix 500: time base can be 0.01s or 1s
  - Base = 0.01, Preset = 500, Time = 5s
- AB RSLogix 5000: time base is *always* 0.001s
  - Base = 0.001, Preset = 500, Time = 0.5s

Timer Variable Warning

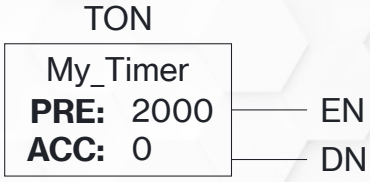
- Timer variables can only be used *once*
- Software will *allow* you to use them more than once, but you will definitely see fun and interesting behavior
- One timer variable per timer instruction

Timers and Counters

Timer On

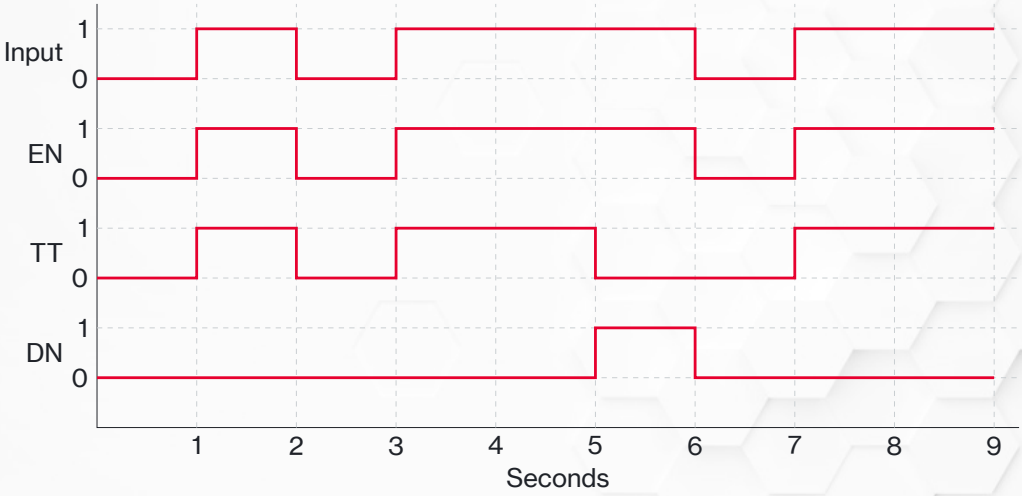
Timer On Instruction

- Starts timing when the preceding logic is true
- Stops timing when the logic goes false
- `ACC` is cleared when the logic goes false



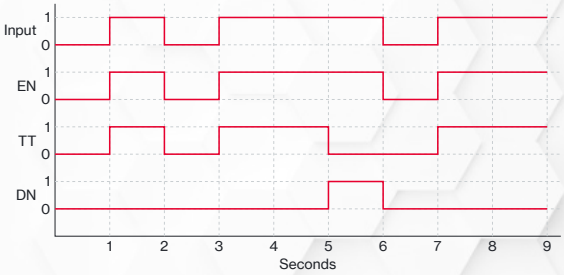
TON Example 1

TON Timing Diagram



TON Timing Step-by-Step (1 of 7)

- At 1 second the input goes true
- The logic is true–EN turns on
- The timer starts timing–TT turns on

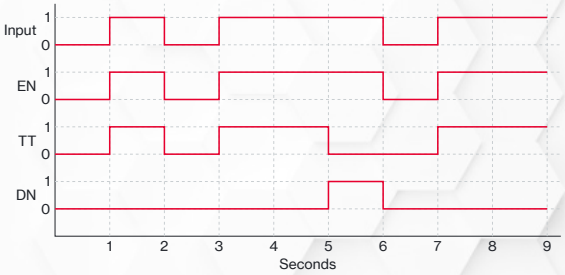


TON Timing Step-by-Step (2 of 7)

While the logic is true `EN` turns on, `TT` turns on, and `ACC` starts counting

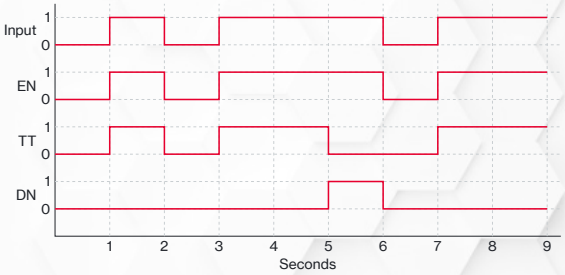
TON Timing Step-by-Step (3 of 7)

- At 2 seconds the input goes false
- The logic is false–EN turns off
- The timer stops timing–TT turns off
- The ACC is automatically cleared
- It wasn't on long enough to reach 2s, DN never turns on



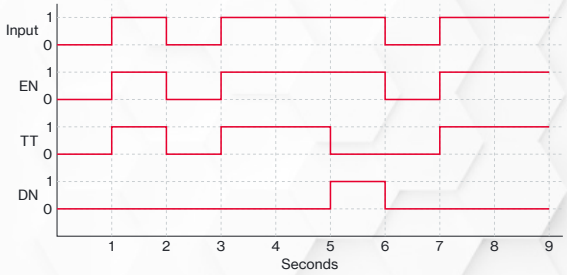
TON Timing Step-by-Step (4 of 7)

- At 3 seconds the input goes true
- The logic is true–EN turns on
- The timer starts timing–TT turns on
- ACC starts counting



TON Timing Step-by-Step (5 of 7)

- At 5 seconds the input is still on
- The logic is true–`EN` stays on
- `ACC` ≥ `PRE`–`DN` turns on
- `ACC` stops counting, holds its value
- It's not actively timing–`TT` turns off

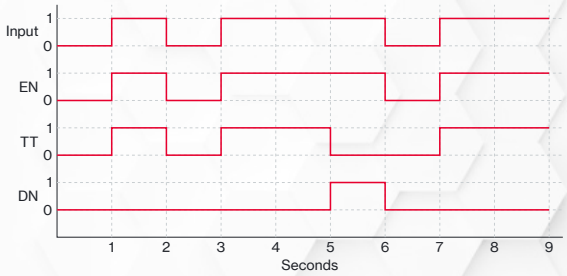


TON Timing Step-by-Step (6 of 7)

At 5 seconds `ACC` ≥ `PRE` so `DN` turns on and `TT` turns off

TON Timing Step-by-Step (7 of 7)

- At 6 seconds the input goes false
- The logic is false–`EN` turns off
- The `ACC` is automatically cleared
- `DN` turns off



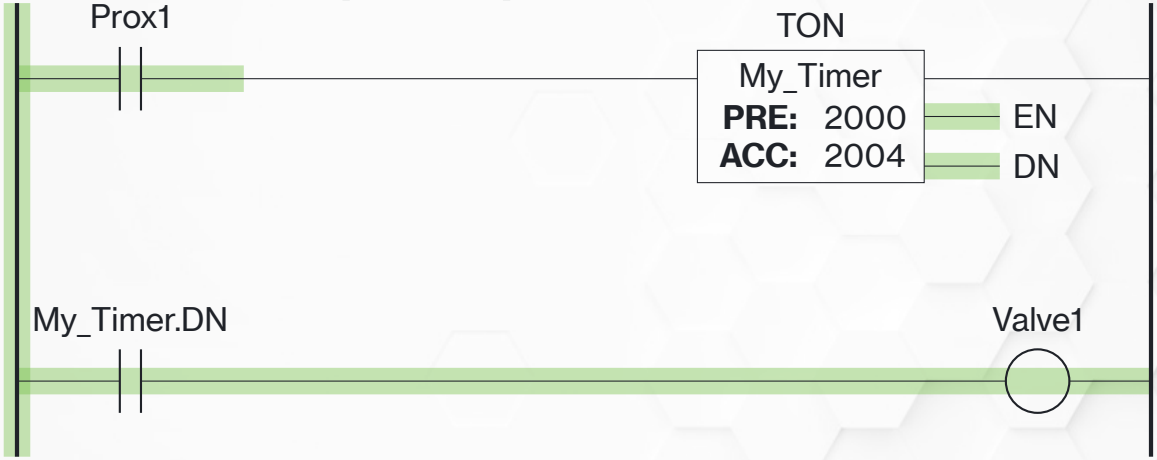
ACC and DN (1 of 3)

- Note the ACC value when DN turns on–this is normal
- The TON *logic* is triggered by the PLC scan
- The TON *timing* happens in a separate processor thread
- The scan checks in every pass to compare the new ACC to PRE

ACC and DN (2 of 3)

- The `ACC` will *always* be slightly bigger than `PRE` when the timer finishes
- You *cannot* do an equal comparison (`ACC=PRE`) because it's *never* true
- That's what `DN` is for

ACC and DN (3 of 3)



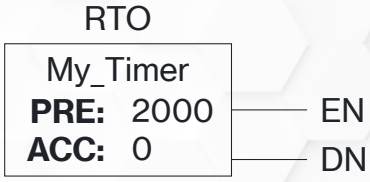
Use `DN` to trigger logic that has to happen when the timer finishes, *not* an equal comparison

Timers and Counters

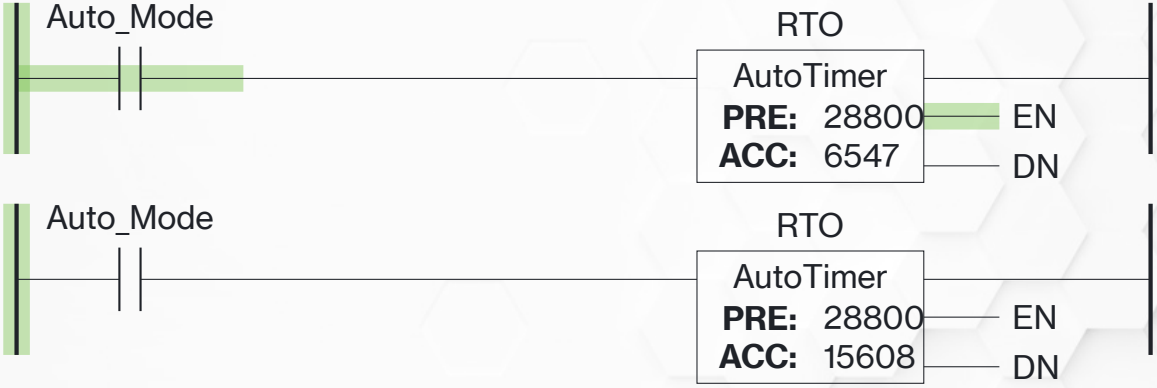
Retentive Timer On

Retentive Timer On (RTO)

- RTO works exactly like TON with *one* difference
- `ACC` *does not* clear when the input goes false
- `ACC` must be explicitly cleared with a RES instruction



RTO Example (1 of 3)

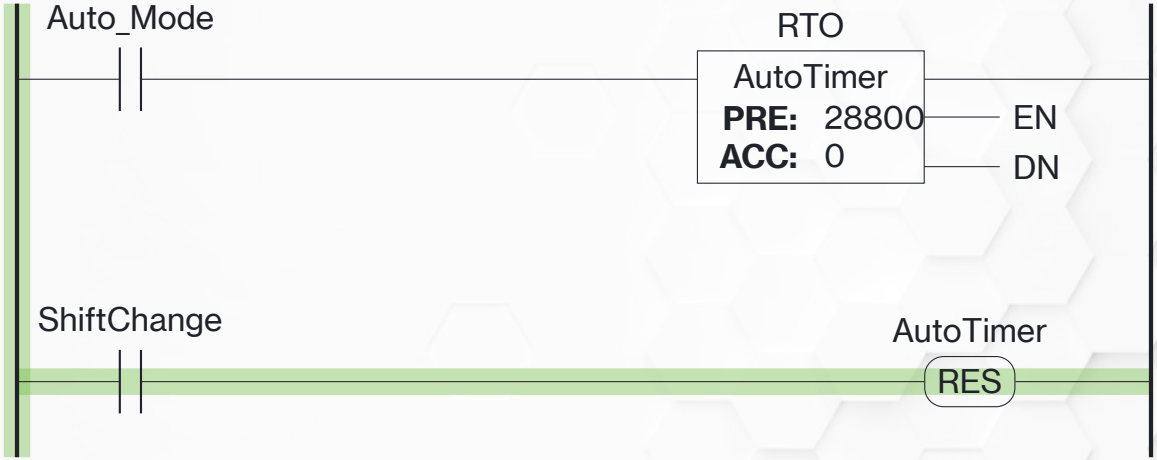


Top: the RTO times as normal when the logic is true. Bottom: When the logic false, the RTO stops timing but retains the ACC.

RTO Example (3 of 3)

When PRE is reached DN turns on

RTO Example (3 of 3)



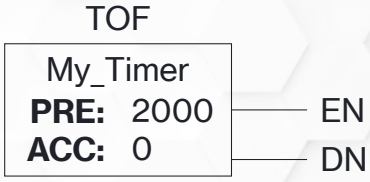
The `ACC` and `DN` will *stay on* until a RES instruction is used on the timer

Timers and Counters

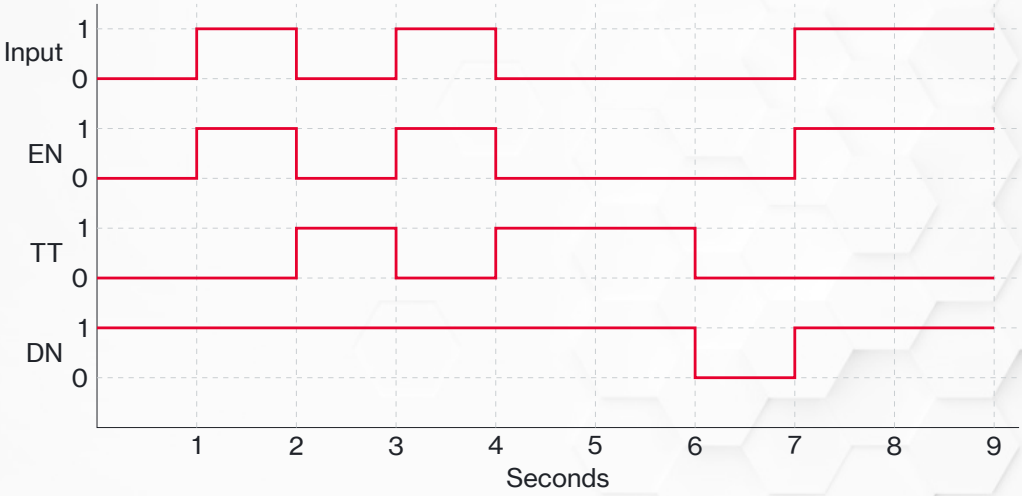
Timer Off

Timer Off Instruction

- Used to keep things running *after* the timer finishes (cooling fans)
- `DN` is on most of the time, it turns *off* when the timer finishes

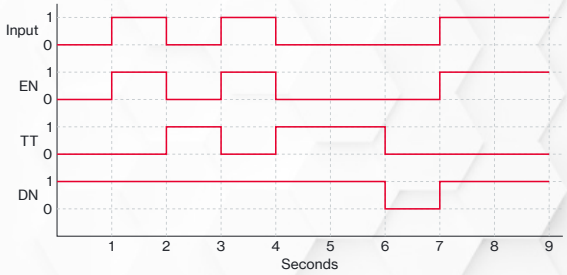


TOF Timing Diagram



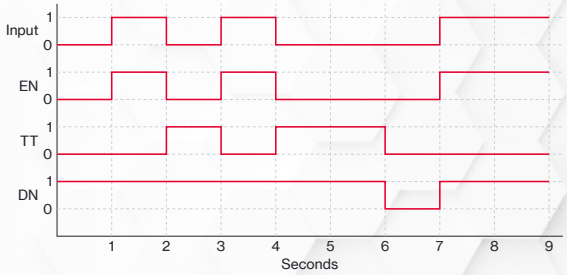
TOF Timing Step-by-Step (1 of 7)

- At 0s the input it off
- `DN` is *already* on



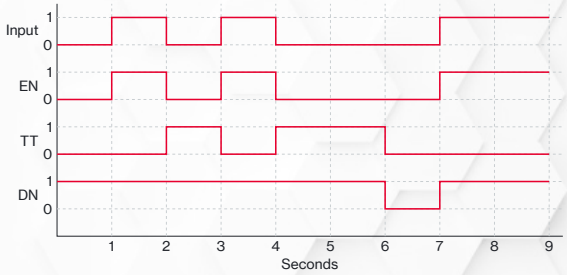
TOF Timing Step-by-Step (2 of 7)

- At 1s the input turns on
- `EN` turns on
- `DN` is *still* on



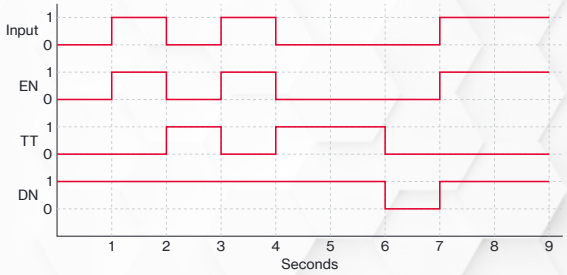
TOF Timing Step-by-Step (3 of 7)

- At 2s the input turns off
- `EN` turns off
- `TT` turns on
- The timer starts timing as soon as the input turns off



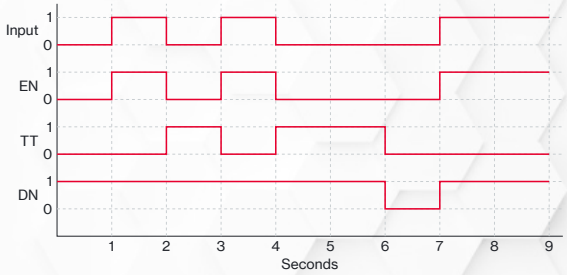
TOF Timing Step-by-Step (4 of 7)

- At 3s the input turns on
- `EN` turns on
- `TT` turns off
- The timer preset hasn't been met, nothing happens with `DN`



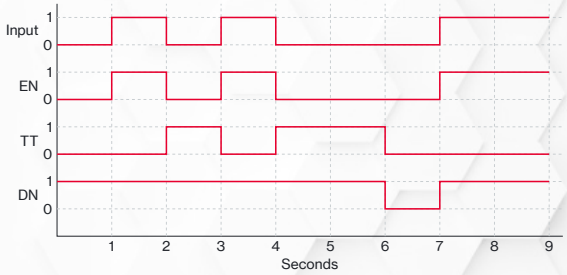
TOF Timing Step-by-Step (5 of 7)

- At 4s the input turns off
- `EN` turns off
- `TT` turns on
- The timer starts timing as soon as the input turns off



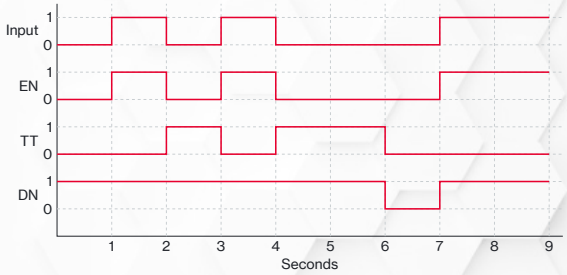
TOF Timing Step-by-Step (6 of 7)

- At 6s we hit the preset
- `TT` turns off–it's not actively timing anymore
- `DN` turns *off*

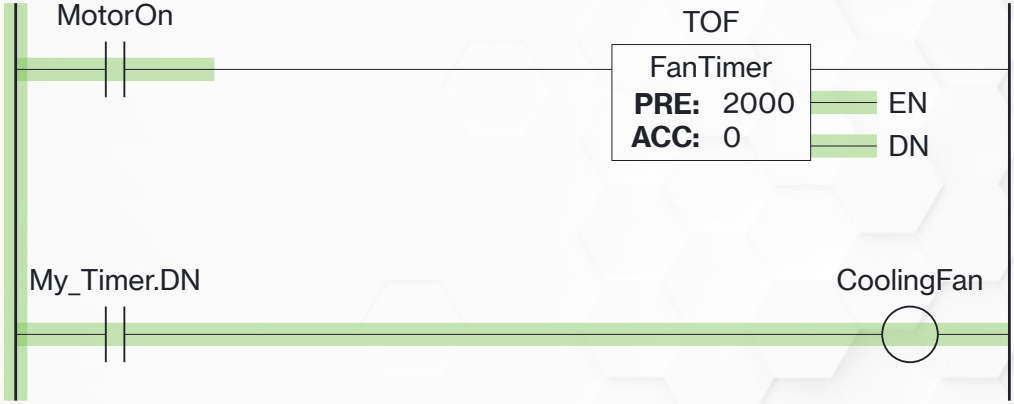


TOF Timing Step-by-Step (7 of 7)

- At 7s the input turns on
- `EN` turns on
- `DN` turns *on*

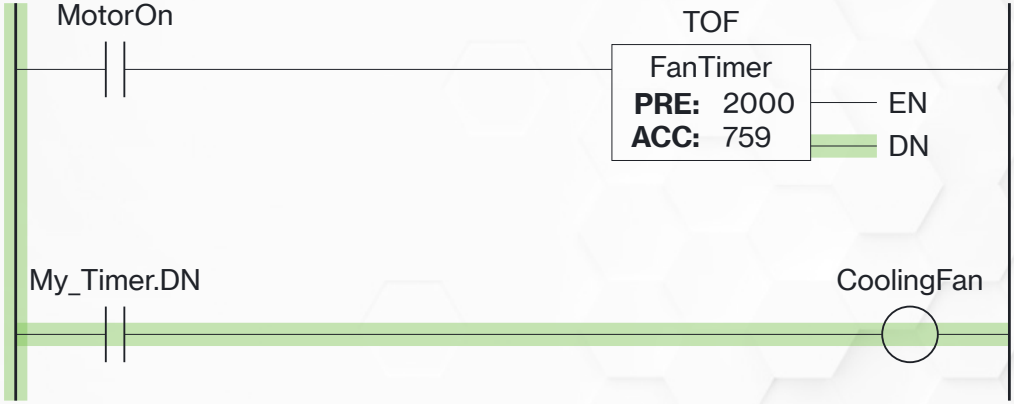


TOF Example (1 of 3)



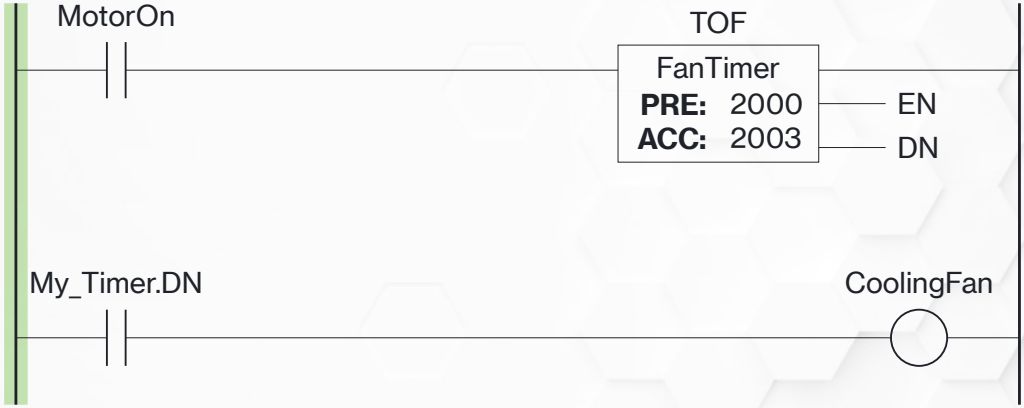
While the input is on, `DN` remains on and timer *is not* timing

TOF Example (2 of 3)



As soon as the input drops out the timer starts timing–EN drops, but DN stays on

TOF Example (3 of 3)



When we hit `PRE`, `DN` drops out

Timers and Counters

Counters

Counter Variables

CU The logic before the `CTU` instruction is true
CD The logic before the `CTD` instruction is true
DN The preset value has been reached
OV The `ACC` has exceeded the DINT limit of 2,147,483,647
UN The `ACC` has exceed the DINT limit of -2,147,483,648
PRE The target count
ACC The current count

Counter Usage

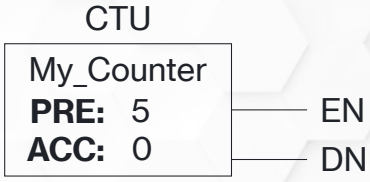
- Unlike timer variables, counter variables can be used *twice*
- Once for a count up (CTU) and once for a count down (CTD)
- Recommendation: put the CTU and CTD *very* close to one another–makes it easier to troubleshoot

Timers and Counters

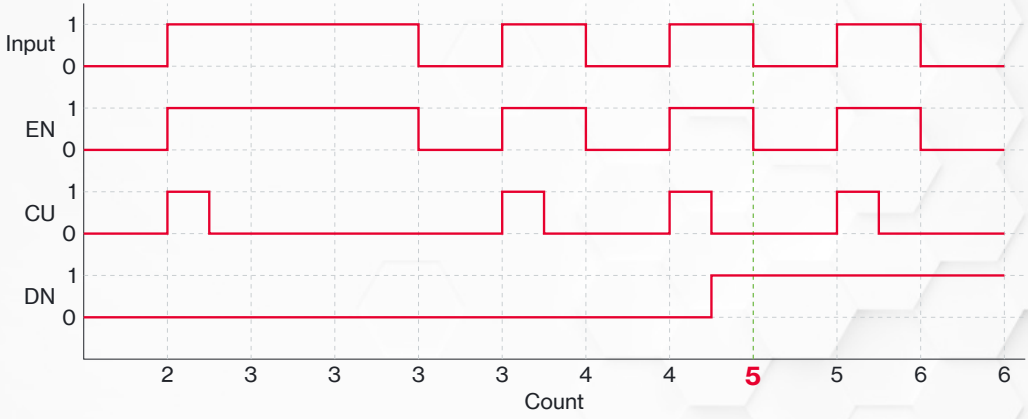
Count Up

Count Up (CTU)

- Adds one to the count when the preceding logic is true
- *ONLY* adds one–the logic must go false in order to count again
- If the counter variable needs to be cleared, must use a RES instruction

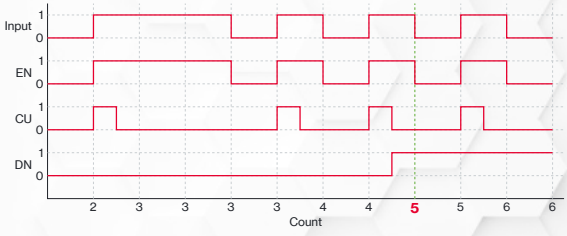


CTU Timing Diagram



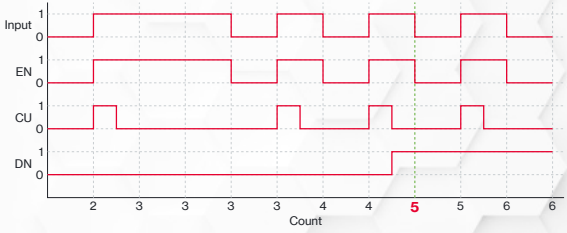
CTU Step-by-Step (1 of 4)

- The input becomes true
- EN turns on
- CU turns on–it only stays on long enough for the count to happen
- The ACC is incremented by 1
- As long as the input stays on CU never pulses, ACC stays where it is



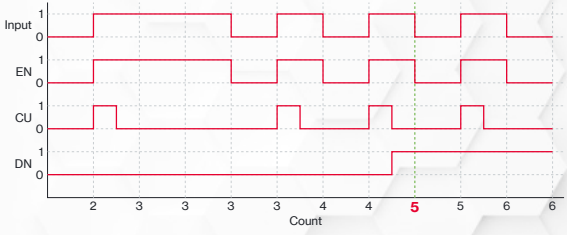
CTU Step-by-Step (2 of 4)

- After several input pulses `ACC = PRE`
- `DN` energizes



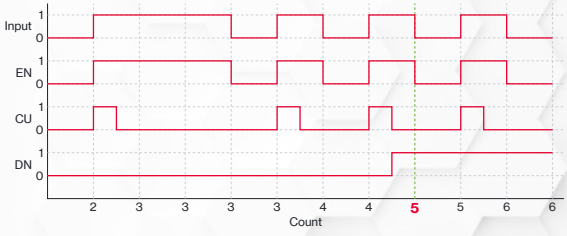
CTU Step-by-Step (3 of 4)

- If left alone, the counter will keep counting every input pulse
- `DN` stays on as long as `ACC ≥ PRE`

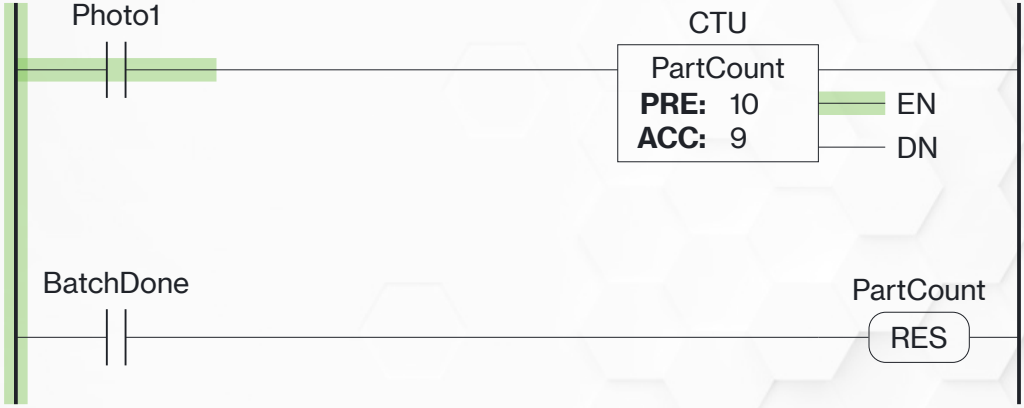


CTU Step-by-Step (4 of 4)

- Eventually the `ACC` will exceed 2,147,483,647 (`DINT` limit)
- At that point `ACC` will roll over to read -2,147,483,648 and count up from there
- `DN` will stay on
- `OV` will turn on

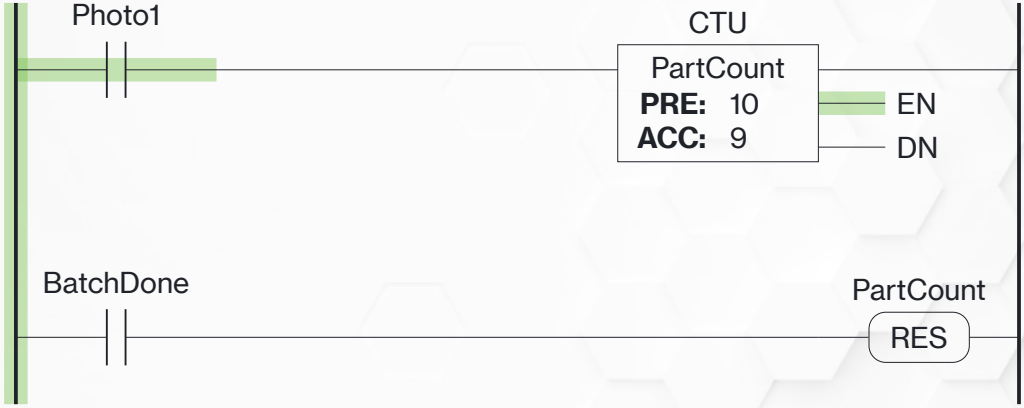


CTU Example 1 (1 of 7)



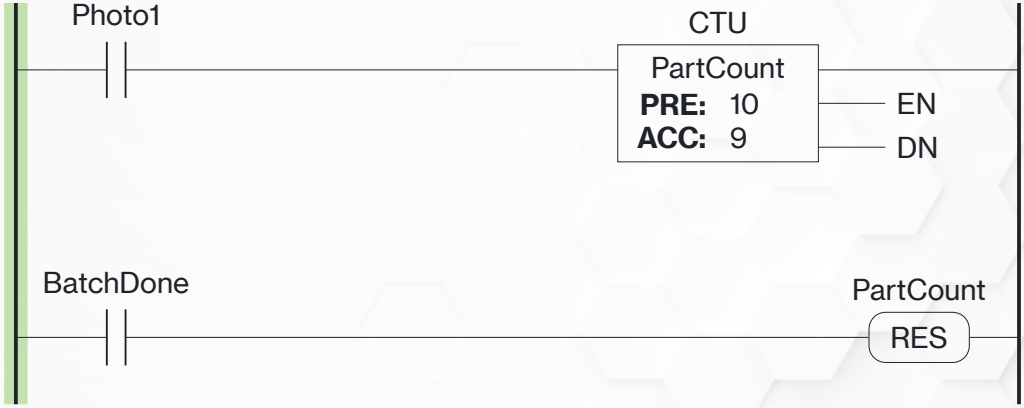
The input turns on, EN turns on, and the count increments by one

CTU Example 1 (2 of 7)



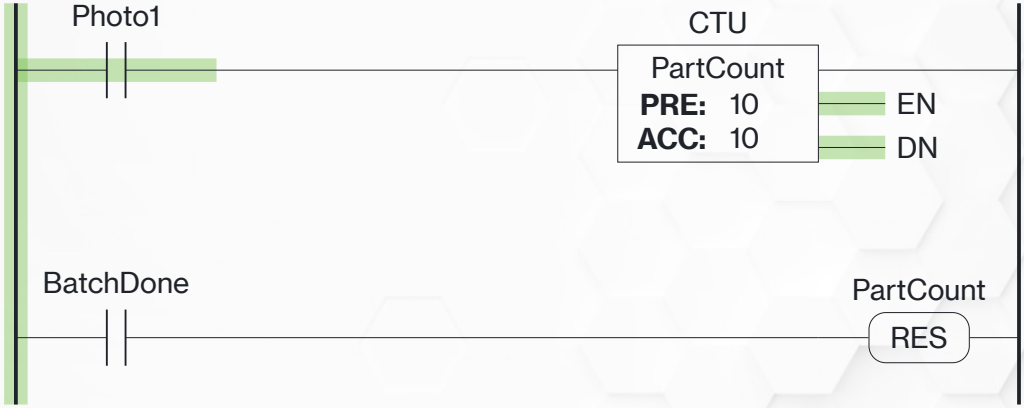
As long as the input *stays* on, `EN` stays on but `ACC` stays where it is

CTU Example 1 (3 of 7)



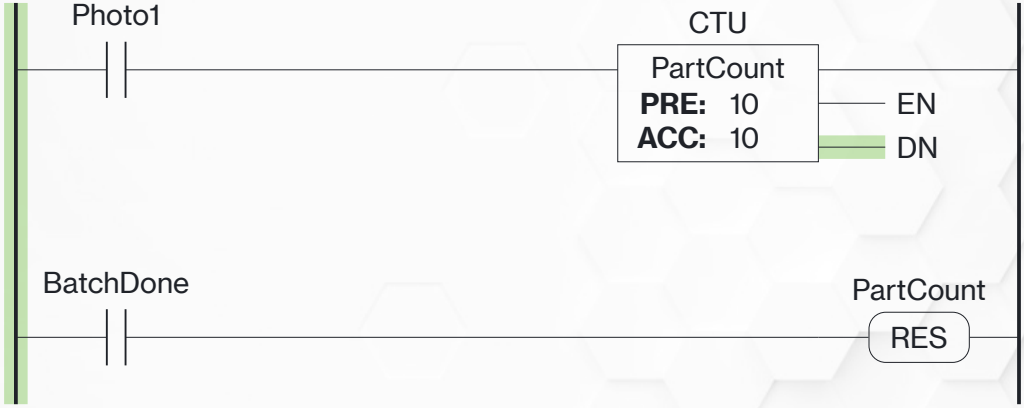
When the input turns off, `EN` turns off

CTU Example 1 (4 of 7)



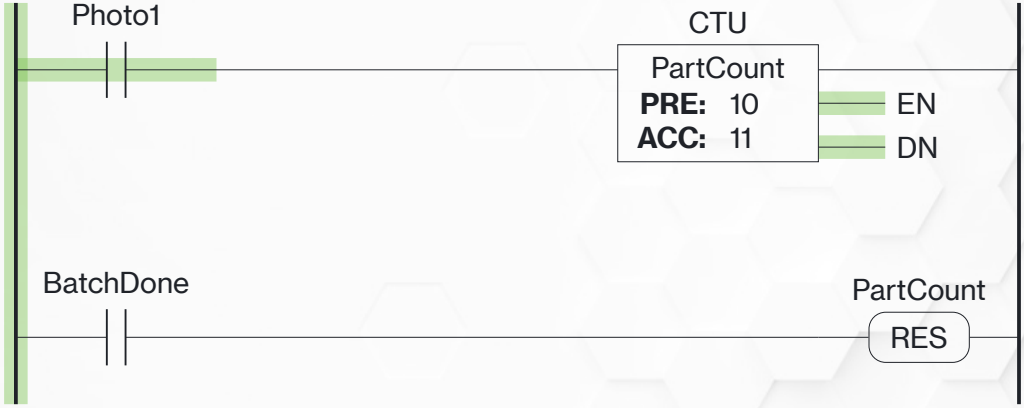
When the input turns on again, `EN` turns on and `ACC` increments. Because `ACC` = `PRE`, `DN` also turns on.

CTU Example 1 (5 of 7)



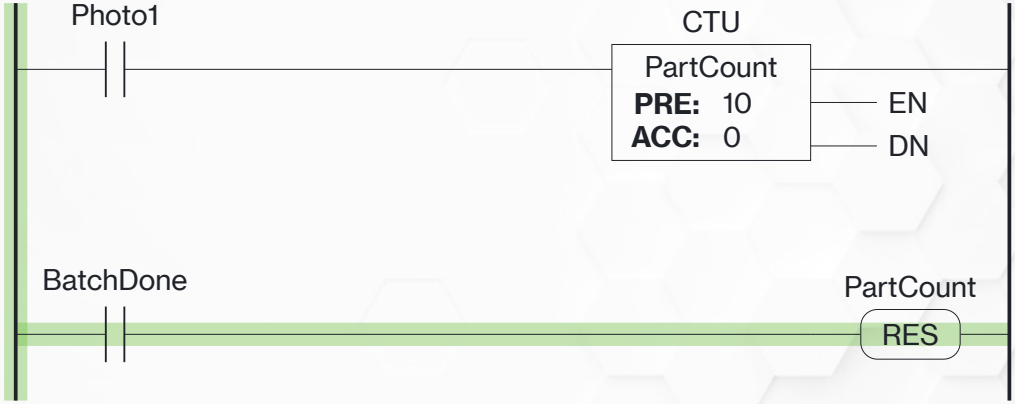
When the input drops out, so does `EN`. But `DN` stays on.

CTU Example 1 (6 of 7)



ACC will keep incrementing with each new input pulse, and DN will stay on

CTU Example 1 (7 of 7)



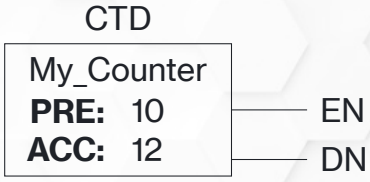
A RES instruction can be used on the counter variable to reset the `ACC` back to 0

Timers and Counters

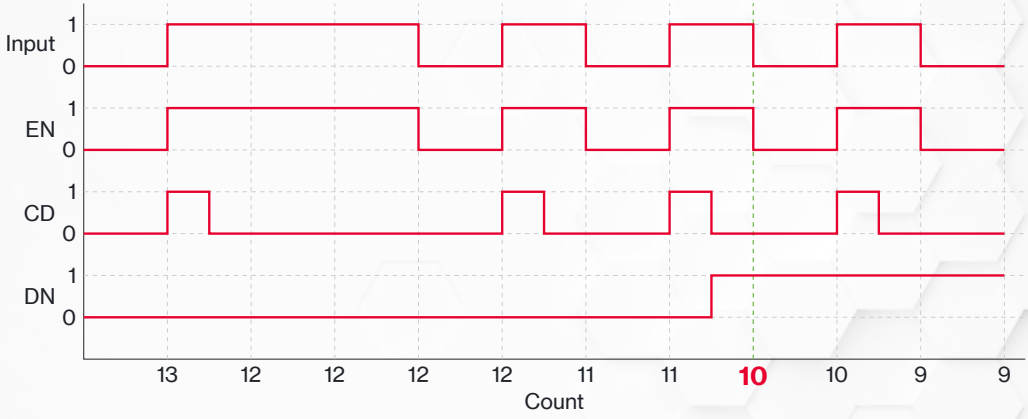
Count Down

Count Down (CTD)

- Subtracts one from the count when the preceding logic is true
- *ONLY* subtracts one–the logic must go false in order to count again
- *DON'T* use a RES instruction to reset the counter...

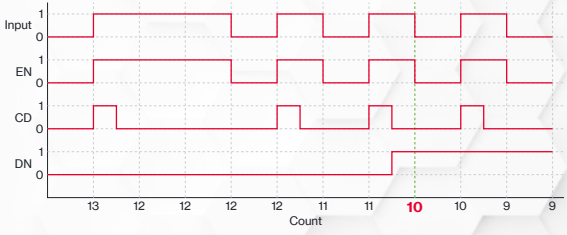


CTD Timing Diagram



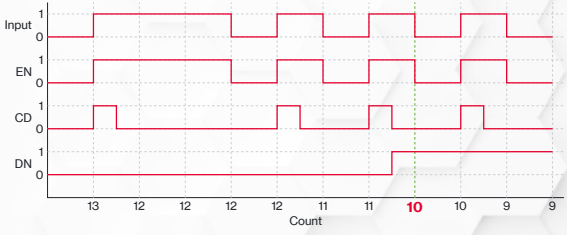
CTD Step-by-Step (1 of 4)

- The input becomes true
- EN turns on
- CD turns on–it only stays on long enough for the count to happen
- The ACC is decremented by 1
- As long as the input stays on CD never pulses, ACC stays where it is



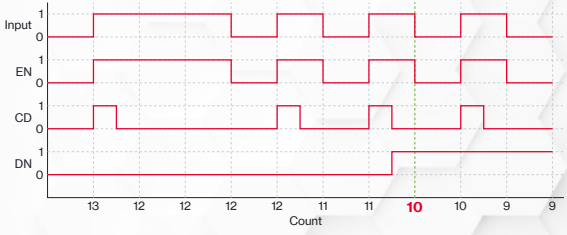
CTD Step-by-Step (2 of 4)

- After several input pulses
  `ACC = PRE`
- `DN` energizes



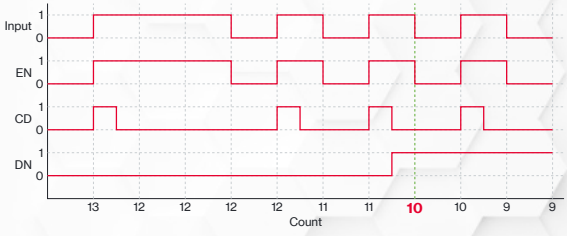
CTD Step-by-Step (3 of 4)

- If left alone, the counter will keep counting every input pulse
- `DN` stays on as long as `ACC` ≤ `PRE`

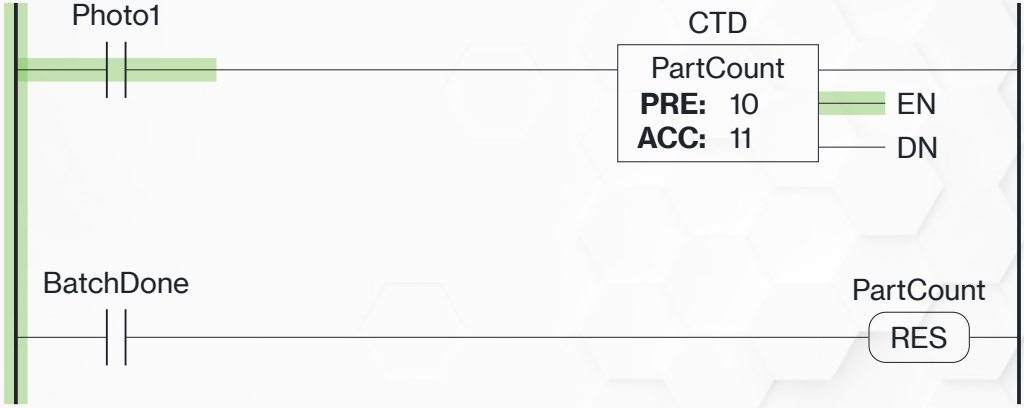


CTD Step-by-Step (4 of 4)

- Eventually the ACC will exceed -2,147,483,648 (DINT limit)
- At that point ACC will roll over to read 2,147,483,647 and count up from there
- DN will stay on
- UN will turn on

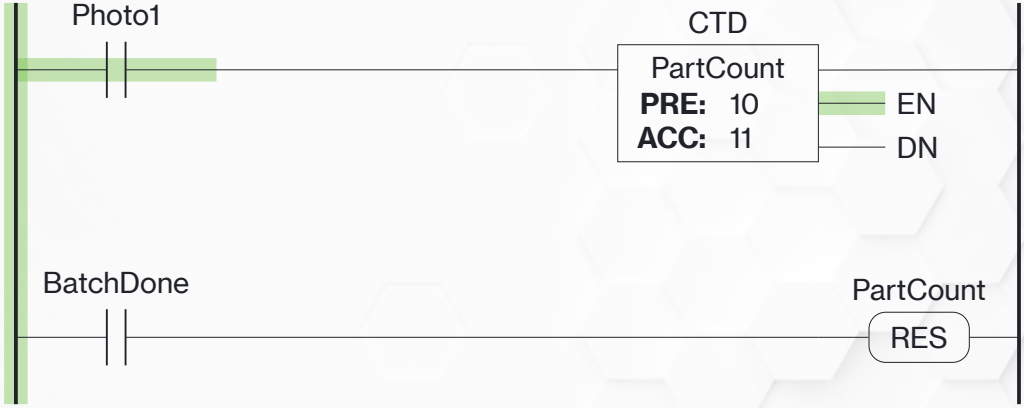


CTD Example 1 (1 of 7)



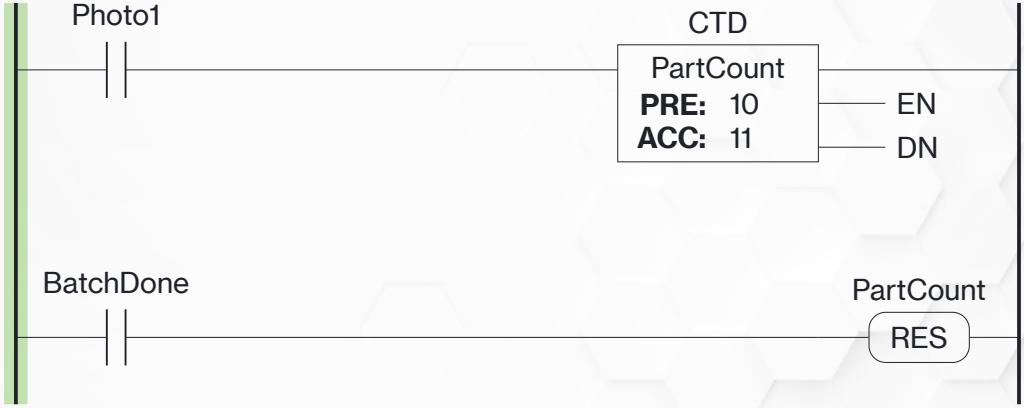
The input turns on, `EN` turns on, and the count decrements by one

CTD Example 1 (2 of 7)



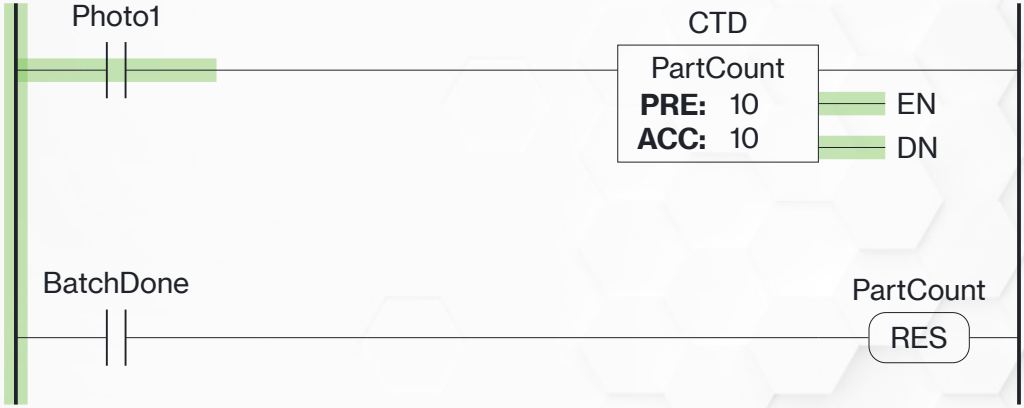
As long as the input *stays* on, `EN` stays on but `ACC` stays where it is

CTD Example 1 (3 of 7)



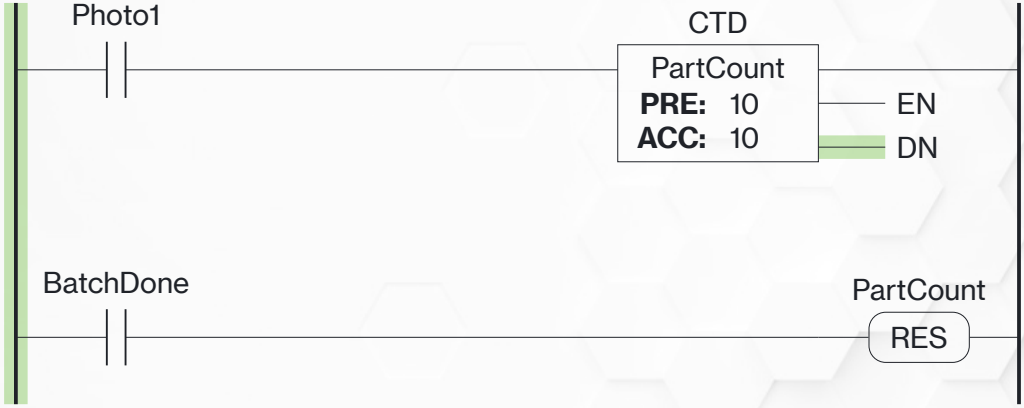
When the input turns off, EN turns off

CTD Example 1 (4 of 7)



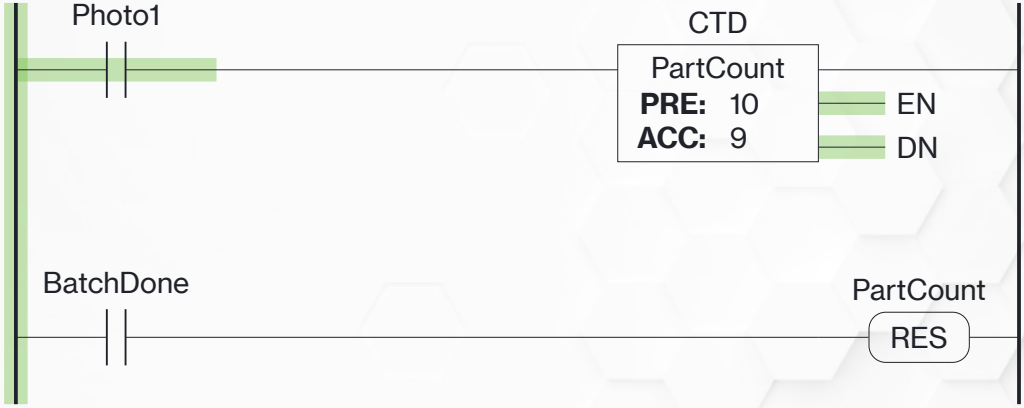
When the input turns on again, `EN` turns on and `ACC` decrements. Because `ACC` = `PRE`, `DN` also turns on.

CTD Example 1 (5 of 7)



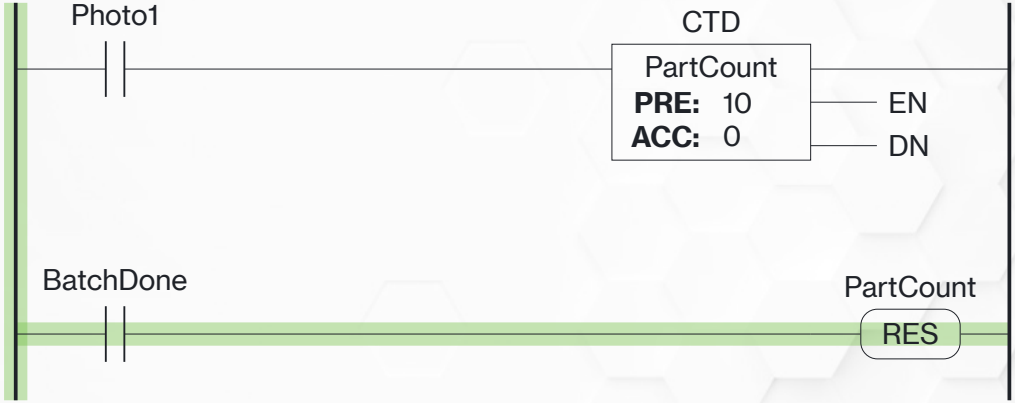
When the input drops out, so does `EN`. But `DN` stays on.

CTD Example 1 (6 of 7)



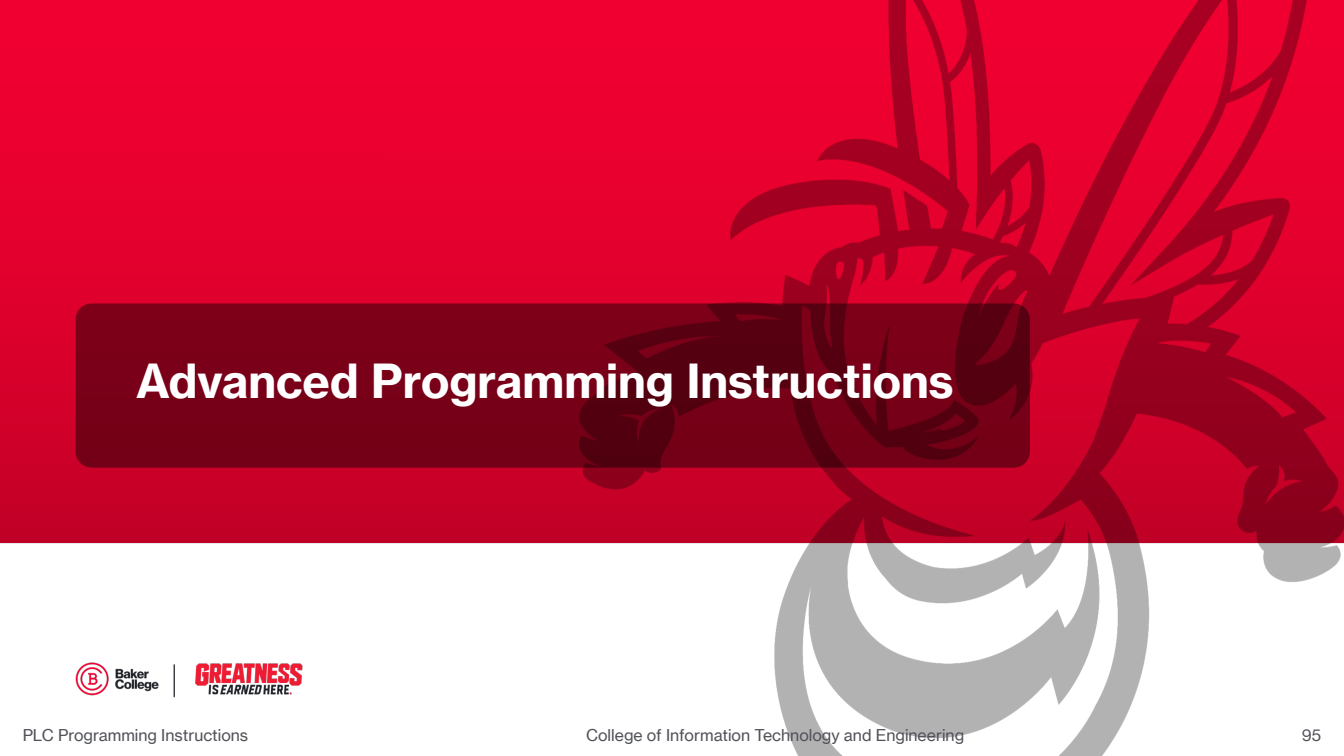
`ACC` will keep decrementing with each new input pulse, and `DN` will stay on

CTD Example 1 (7 of 7)



A RES instruction sets `ACC` to 0–probably not a good thing for a count down

Advanced Programming Instructions



Advanced Programming Instructions

- PLCs are computers–capable of most functions a computer can perform
- In our industrial world, we're mostly going to focus on comparison and math
- Comparison instructions are *decision* instructions–if they evaluate as true "power" flows through
- Math instructions are *action* instruction–they go at the end of the rung (output) at only evaluate if the rest of the rung is true

Advanced Programming Instructions

Compare Instructions

Compare Instruction Naming Convention

- In Logix Designer v36 Rockwell renamed most of these instructions
- The renaming aligns AB with IEC 61131-3
- Older programs should automatically convert instruction names when updated to v36
- Older *programmers* need to learn new muscle memory when typing instruction logic

Compare Instructions

EQ (*A* = *B*)

Src A:	Number1 10
Src B:	Number2 5

GT (*A* > *B*)

Src A:	Number1 10
Src B:	Number2 2

LT (*A* < *B*)

Src A:	Number1 10
Src B:	Number2 2

GE (*A* ≥ *B*)

Src A:	Number1 10
Src B:	Number2 2

LE (*A* ≤ *B*)

Src A:	Number1 10
Src B:	Number2 3

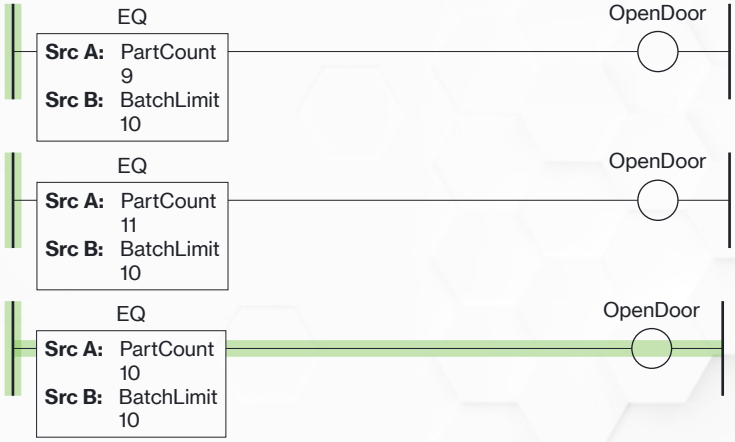
LIMIT

Low:	Number1 10
Test:	Number2 3
High:	Number3 20

Compare Instruction Arguments

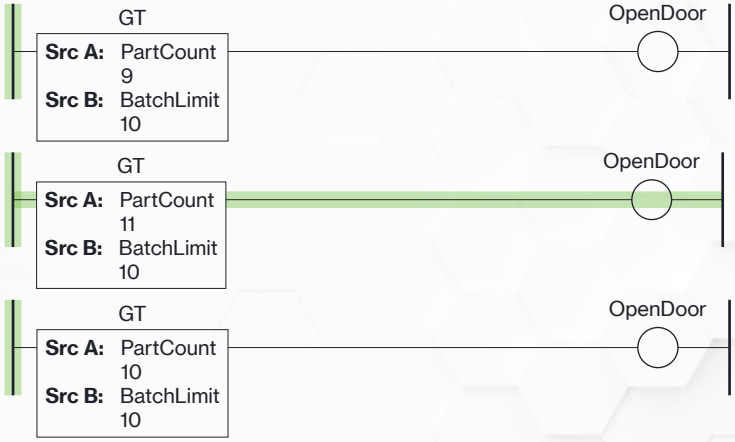
- With the exception of LIMIT, the compare instructions accept two arguments–the two numbers that will be compared
- Source A and Source B can be either fixed or variable
- Standard programming convention–Source A should be the variable, Source B should be the fixed comparison
    - This means you might need to use a LT instead of a GT, depending on the need

Equal Instruction



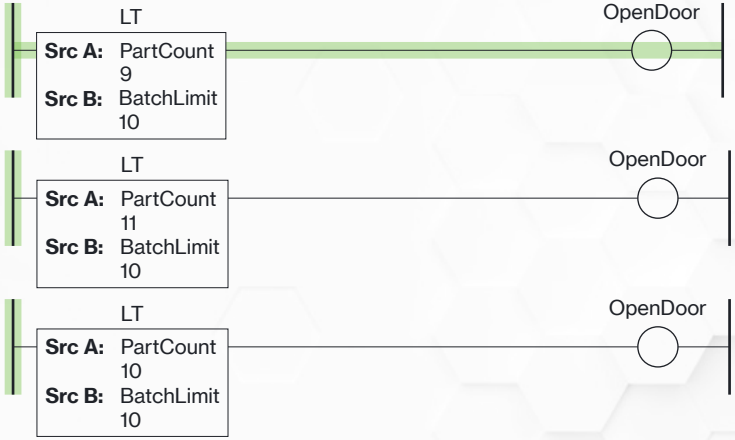
The EQ instruction is true when Source A is *exactly* equal to Source B. Take care with things like timers, analog inputs, etc.–they are *rarely* exactly equal.

Greater Than Instruction



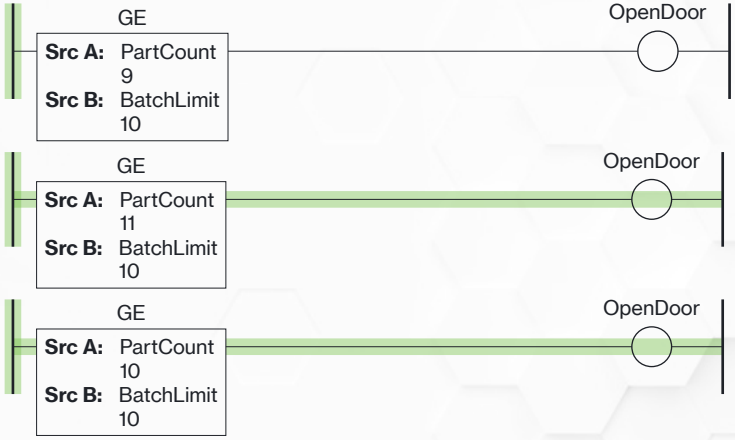
The GT instruction is true when Source A is greater than Source B.

Less Than Instruction



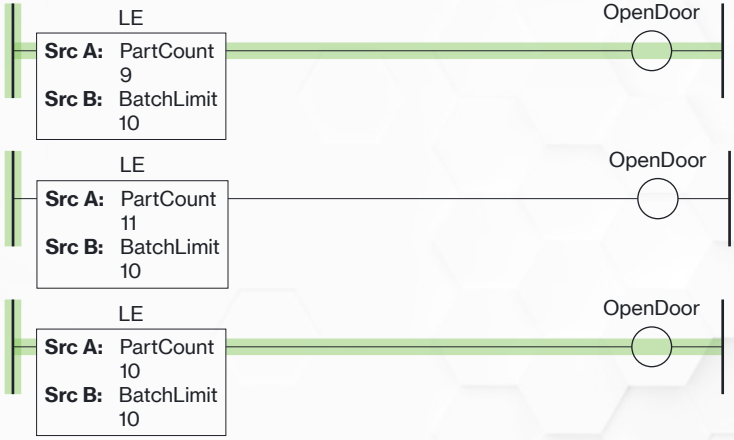
The LT instruction is true when Source A is less than Source B.

Greater Than or Equal Instruction



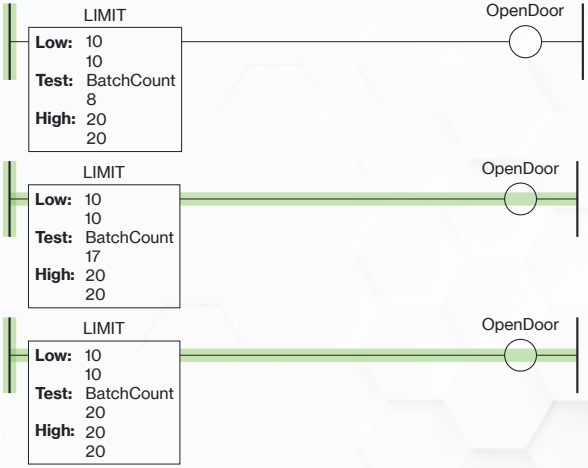
The GE instruction is true when Source A is greater than *or* equal to Source B.

Less Than or Equal Instruction



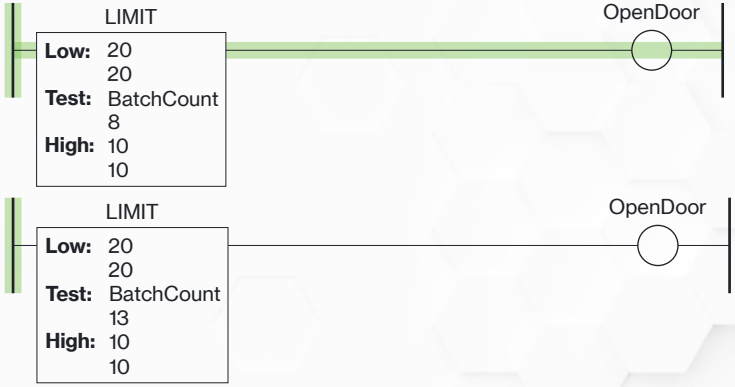
The LE instruction is true when Source A is less than *or* equal to Source B.

Limit Instruction



The LIMIT instruction is true when the test value falls between the high and low limits, inclusive.

Reverse Limit

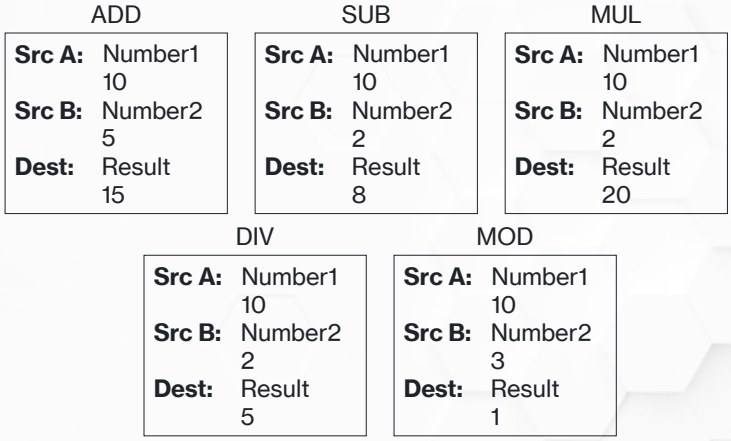


By placing the higher value in the Low field and the lower value in the High field, LIMIT checks if the test is *outside* the limits.

Advanced Programming Instructions

Math Instructions

Math Instructions



There are more math instructions, but these are the most common

Math Instruction Arguments

- The instructions on the previous slide accept 3 arguments: Source A, Source B, and Destination
- Source A and Source B can be either a variable (tag) or a fixed number
- Destination *must* be a variable–the instruction will store the result here

Argument Example

All Fixed

ADD

Src A:	10
	10
Src B:	1
	1
Dest:	Result
	11

One Variable

ADD

Src A:	Count
	11
Src B:	1
	1
Dest:	Result
	12

All Variable

ADD

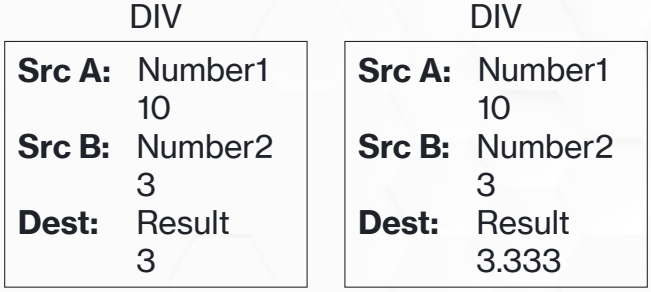
Src A:	Count
	11
Src B:	Increment
	4
Dest:	Result
	15

The sources can be fixed numbers or variables.

Math Instruction Data Types

- Math instructions work on all numeric data types
- In Allen-Bradley, data types *do not* need to match
  - Some brands *do* force them to match
- Be careful–if the math would produce a decimal, but the variable is an integer all decimal points will be chopped off
- Can be useful if you don't want the decimals

Floating Point Example

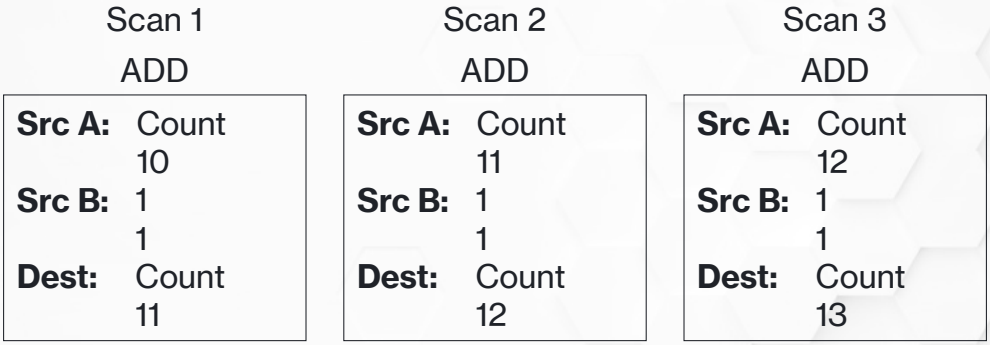


Left: Result is defined as a `DINT`, all decimals are truncated. Right: Result is defined as a `REAL`, decimal points are preserved.

Math Instruction Behavior

- Source A comes first in the operation–Source A minus Source B, Source A divided by Source B, etc
- The destination variable *can be* the same variable as Source A–useful for a running count

Count Example



Setting the destination to the same variable as source A stores the result back in source A

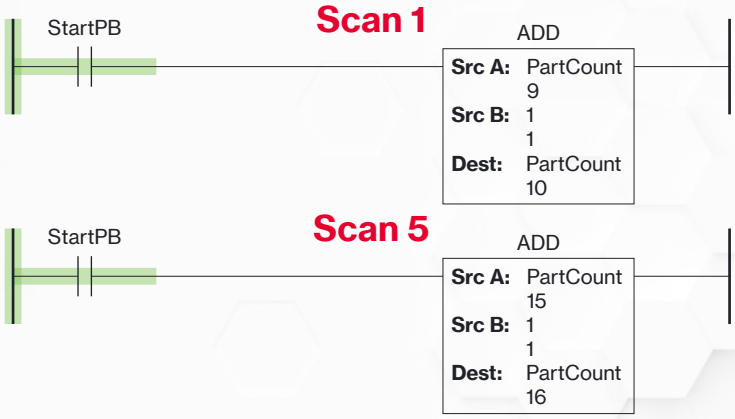
Protecting Math

- Math instructions can lead to some tricky program behavior–even processor hard faults
- Care must be taken to *protect* the math and prevent undesired operation

Protecting Addition (1 of 3)

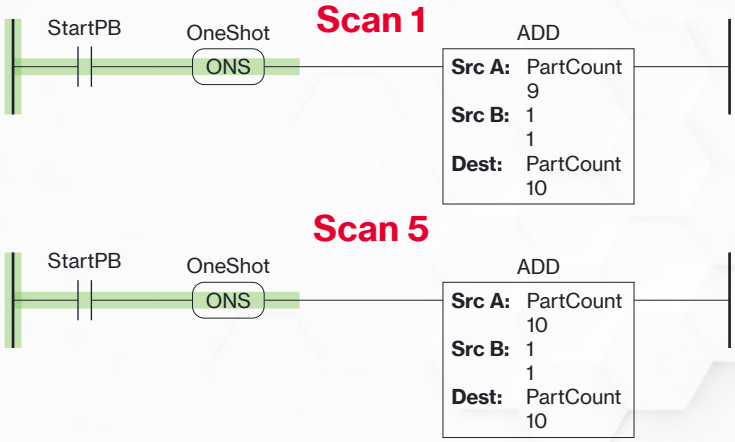
- Math instructions execute *every scan* while the preceding logic is true
- This can be especially dangerous when using an ADD as a counter
- Remember, if an input is human actuated it *will* be on for more than one scan–leads to more than one count

Protecting Addition (2 of 3)



An ADD set up as a counter is going to count each *scan* the logic is true, not necessarily each toggle of the logic

Protecting Addition (3 of 3)



ALWAYS add a one-shot to guarantee the addition only executes once per logic toggleand prevent runaway

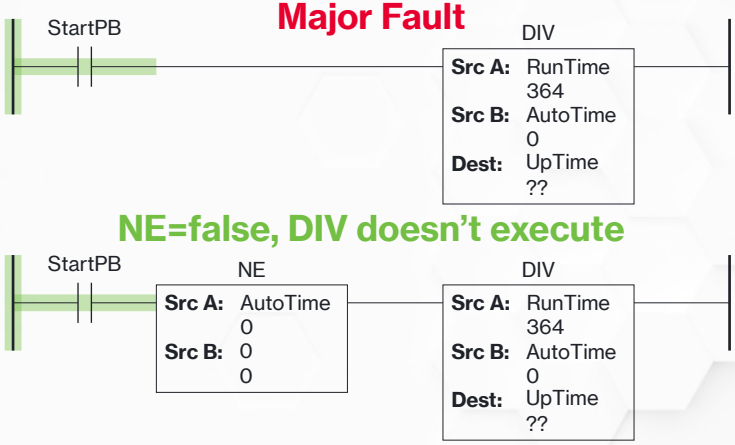
Protecting Division (1 of 3)

There is also a major potential problem with division–what is it?

Protecting Division (2 of 3)

- “Divide by Zero” is a problem in *any* computer system
- In PLCs, it will cause a major fault–which immediately kills all program execution
- It’s also fairly simple to protect against–how?

Protecting Division (3 of 3)



Using the division's Source B in a not equal to zero instruction prevents the division from executing a divide by zero

Questions?

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